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Your Roll No.....

Sr. No. of Question Paper : 5849

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Unique Paper Code : 2352013503

Name of the Paper : Partial Differential Equations

**Name of the Course : B.Sc. (H) Mathematics –
DSC-15**

Semester : V

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All questions are compulsory and are of 15 marks each.
3. Attempt any two parts from each Questions. Each part is of 7.5 marks.

1. (a) Define characteristic curves of a first order partial differential equation. Find the partial differential equation arising from the two-parameter family of curves :

$$u - ax - by - ab = 0.$$

- (b) Determine the integral surfaces of the equation:

$$x(y^2 + u)u_x - y(x^2 + u)u_y = (x^2 - y^2)u; \text{ with the data } x + y = 0 \text{ and } u = 1.$$

(c) Find the solution of the initial value system:

$$u_t - 2uu_x = v - x,$$

$$v_t + cv_x = 0,$$

$$u(x, 0) = x, v(x, 0) = x.$$

2. (a) Reduce the given equation into canonical form and find the general solution x :

$$u_x + xu_y = y.$$

- (b) Using $v = \ln u$ and $v(x, y) = f(x) + g(y)$, solve the equation:

$$x^2u_x^2 + y^2u_y^2 = u^2.$$

- (c) Find a complete integral of the following equation by using Charpit's method :

$$z^2 = pqxy.$$

3. (a) Derive the three-dimensional Laplace equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0,$$

where V is Gravitational potential at any point.

- (b) Determine the region of the following equation, and transform it to canonical form:

$$x^2u_{xx} - 2xyu_{xy} + y^2u_{yy} = e^x.$$

- (c) Derive the equation of traffic flow :

$$\frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} = 0,$$

where $\rho(x, t)$ is traffic density and $q(x, t)$ is traffic flow at any time t .

4. (a) Find the general solution of the equation:

$$yu_{xx} + 3yu_{xy} + 3u_x = 0, \quad y \neq 0.$$

- (b) Reduce the following equation into canonical form :

$$u_{xx} - 2\cos x u_{xy} + (1 + \cos^2 x)u_{yy} + u = 0.$$

- (c) Find the general solution of the equation:

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial^2 z}{\partial y^2} = e^{x+y}.$$

5. (a) Determine the solution of the initial-value problem for an infinite string with initial condition:

$$\begin{aligned} u_{tt} - u_{xx} &= 0, \quad x \in \mathbb{R}, \quad t > 0, \\ u(x, 0) &= \sin 3x, \quad x \in \mathbb{R}, \\ u_t(x, 0) &= \cos 3x, \quad x \in \mathbb{R}. \end{aligned}$$

- (b) Find the solution of the initial-value problem for a semi-infinite string with fixed end:

$$\begin{aligned} u_{tt} - 16u_{xx} &= 0, \quad 0 < x < \infty, \quad t > 0, \\ u(x, 0) &= \sin x, \quad 0 \leq x < \infty, \\ u_t(x, 0) &= x^2, \quad 0 \leq x < \infty, \\ u(0, t) &= 0, \quad 0 \leq t < \infty. \end{aligned}$$

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- (c) Determine the solution of wave equation of semi-infinite string with free end for non-homogeneous boundary value problem:

$$\begin{aligned} u_{tt} - 16u_{xx} &= 0, & x > 0, & \quad t > 0, \\ u(x, 0) &= \cos x, u_t(x, 0) = 2, & x \geq 0, \\ u_x(0, t) &= 1 - t, & t \geq 0. \end{aligned}$$

6. (a) Obtain the solution for a semi-infinite string with free end:

$$\begin{aligned} u_{tt} &= c^2 u_{xx}, & 0 < x < \infty, & \quad t > 0, \\ u(x, 0) &= f(x), & 0 \leq x < \infty, \\ u_t(x, 0) &= g(x), & 0 \leq x < \infty, \\ u_x(0, t) &= 0, & 0 \leq t < \infty. \end{aligned}$$

- (b) Determine the solution of wave equation for non-homogeneous boundary value problem of semi-infinite string with fixed end:

$$\begin{aligned} u_{tt} - 4u_{xx} &= 0, & x > 0, & \quad t > 0, \\ u(x, 0) &= \log(1 + x), & x \geq 0, \\ u_t(x, 0) &= 1, & x \geq 0, \\ u(0, t) &= t, & t \geq 0. \end{aligned}$$

- (c) Determine the solution of Cauchy problem for non-homogeneous wave equation:

$$u_{tt} = u_{xx} - 2, \quad x \in \mathbb{R}, \quad t > 0,$$

with initial condition $u(x, 0) = \sin x$, $u_t(x, 0) = x$, $x \in \mathbb{R}$.